SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.*(BL3)*

Assessment Tool Weightage: 35%

QUESTIONS: 3 **Duration : 1 Hour**
Total Marks:30

Annexure A

Constraint-Based Problem Solving

Code of Conduct:

I Praveen J (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Constraint-Based Problem Solving

1. A hospital needs to assign 3 doctors (D1, D2, D3) to 3 shifts (Morning, Afternoon, Night) such that:

   Constraints:

   i. D1 cannot work Night shift
   ii. D2 must work before D3 (Morning < Afternoon < Night)
   iii. Only one doctor per shift
   iv. D3 cannot work Morning
   v. All doctors must be assigned exactly one shift

   Apply Backtracking Search to find a valid assignment with all intermediate steps and constraint checks. State the final valid schedule

   Let the shifts be ordered **Morning < Afternoon < Night**.

   1. Assign **D1**. D1 cannot work Night, so try Morning first.
   2. Assign **D2**. D2 must work before D3, so D2 cannot be Morning if D3 still needs a later shift. Try Afternoon.
   3. Assign **D3**. D3 cannot work Morning, and must work after D2. Night is valid.
   4. Check constraints:
      - D1 is not on Night.
      - D2 (Afternoon) is before D3 (Night).
      - Each shift has one doctor.
      - D3 is not on Morning.
      - Each doctor has exactly one shift.

   Final valid schedule: Morning — D1; Afternoon — D2; Night — D3.

2. A robot operates in a grid environment (5×5). It must move from Start (S) to Goal (G). Obstacles block certain paths.

   Constraints:

   i. Movement allowed: Up, Down, Left, Right
   ii. Cost of each move = 1
   iii. Cannot pass through obstacles
   iv. Use Manhattan Distance heuristic

   Using above constraints and BFS Search Algorithm Show step-by-step node expansion and priority queue updates and determine the optimal path and cost**.**

Assume **S = (1,1)**, **G = (5,5)**, and obstacles at **(2,2), (2,3), (4,3), (4,4)**. Coordinates are **(row, column)**. BFS explores neighbors in the order **Right, Down, Left, Up**. Its queue is FIFO; the Manhattan distance is shown for reference but does not control BFS ordering.

Step	Node expanded	Newly discovered nodes (Manhattan distance to G)	Queue after expansion
1	(1,1)	(1,2) [7], (2,1) [7]	(1,2), (2,1)
2	(1,2)	(1,3) [6]	(2,1), (1,3)
3	(2,1)	(3,1) [6]	(1,3), (3,1)
4	(1,3)	(1,4) [5], (2,3) blocked	(3,1), (1,4)
5	(3,1)	(3,2) [5], (4,1) [5]	(1,4), (3,2), (4,1)
6	(1,4)	(1,5) [4], (2,4) [5]	(3,2), (4,1), (1,5), (2,4)
7	(3,2)	(3,3) [4], (4,2) [4]	(4,1), (1,5), (2,4), (3,3), (4,2)
8	(4,1)	(5,1) [4]	(1,5), (2,4), (3,3), (4,2), (5,1)
9	(1,5)	(2,5) [3]	(2,4), (3,3), (4,2), (5,1), (2,5)
10	(2,4)	—	(3,3), (4,2), (5,1), (2,5)
11	(3,3)	—	(4,2), (5,1), (2,5)
12	(4,2)	—	(5,1), (2,5)
13	(5,1)	(5,2) [3]	(2,5), (5,2)
14	(2,5)	(3,5) [2]	(5,2), (3,5)
15	(5,2)	(5,3) [2]	(3,5), (5,3)
16	(3,5)	(4,5) [2]	(5,3), (4,5)
17	(5,3)	—	(4,5)
18	(4,5)	(5,5) [0]	(5,5)
19	(5,5)	Goal reached	—

Optimal path found by BFS:
(1,1) → (1,2) → (1,3) → (1,4) → (1,5) → (2,5) → (3,5) → (4,5) → (5,5)

Path cost: 8 moves. BFS finds a shortest path when every move has equal cost.

3. An autonomous rescue robot is deployed in a disaster-affected building to locate and rescue survivors. The robot operates in a partially observable environment where some paths are blocked, and it must make decisions with limited information.

The building is represented as a grid of rooms. The robot starts at position S and must reach a survivor located at G.

Constraints:

- The robot can move up, down, left, right

- Some cells are blocked (obstacles) and cannot be traversed
- The robot has limited sensing capability (can only observe adjacent cells)
- Each movement has a cost of 1, but entering risky zones has an additional cost of 2
- The robot must minimize total cost while ensuring safe navigation
- The robot must update its knowledge dynamically (online search scenario)

Tasks:
a) Analyze all the given constraints and explain how they affect the robot's decision-making process.
b) Develop a solution approach using an appropriate search strategy (e.g., Uniform Cost Search / Online Search Agent). Clearly explain the steps involved.
c) Demonstrate how the robot applies AI concepts such as:
- Intelligent agents
- Rationality
- Environment type

d) Justify why your chosen approach is the most suitable for solving this problem under the given constraints.

a) How the constraints affect decisions
- **Four-way movement:** From each room, the robot can consider only the adjacent room above, below, left, or right.
- **Obstacles:** Blocked rooms are not valid destinations, so the robot must avoid them.
- **Limited sensing:** The robot sees only adjacent rooms. It cannot plan using a complete map and must gather information as it moves.
- **Movement and risk costs:** Every move costs 1. Entering a risky room adds 2, so that move costs 3 in total. The robot should prefer a safer route when it has a lower total cost.
- **Dynamic knowledge:** As the robot senses rooms, it updates its map. New information may change which route appears best, so it must replan when needed.
- **Safety:** A low-cost route is not automatically acceptable if it creates an unacceptable safety risk. The robot should treat known obstacles as forbidden and account for risky rooms in its path costs.

b) Search approach: Online agent with Uniform Cost Search
Uniform Cost Search (UCS) is suitable for comparing routes with different costs. It expands the route with the lowest accumulated cost so far, rather than simply choosing the route with the fewest moves.
1. The robot starts at **S** with a known map that may be incomplete.
2. It senses the four adjacent rooms and marks each as open, blocked, or risky if that information is available.
3. It assigns each valid move a cost: **1** for an ordinary room, or **3** for entering a risky room.
4. It uses UCS on the currently known map to choose the lowest-cost route toward **G**.
5. It moves one step along that route and senses again.
6. It updates its map with the new observations.
7. If the new information blocks the planned route or reveals a cheaper or safer route, it runs UCS again.
8. It repeats this process until it reaches **G**, while avoiding blocked rooms and accounting for risk costs.

The priority queue in UCS is ordered by **total cost from S so far**. For example, a route costing 4 is considered before one costing 6, even if the latter uses fewer steps.

c) AI concepts

- **Intelligent agent:** The robot senses its surroundings, updates its internal map, selects an action, and moves toward the survivor.
- **Rationality:** A rational robot chooses the action that best supports its goal, using the information currently available. It minimizes expected route cost while respecting safety constraints. Because its information is incomplete, it may need to revise its decisions after sensing more.
- **Environment type:** The building is:
    - **Partially observable:** The robot sees only adjacent rooms.
    - **Sequential:** Each move affects the robot's later options and decisions.
    - **Dynamic from the agent's perspective:** Its knowledge changes as it explores; the actual building may also change if hazards shift.
    - **Discrete:** Rooms and movement choices are distinct.
    - **Stochastic or uncertain:** The state of unseen rooms is unknown; outcomes may also be uncertain if the building changes.

d) Why this approach is suitable

UCS accounts for both ordinary movement and the extra cost of risky rooms, so it can find the least-cost route on the map available to the robot. The online sensing and replanning process addresses partial observability: the robot does not assume it knows the entire building in advance. Together, these methods let it update its route as it discovers obstacles and hazards.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided